Program:

app.module.ts :

```
import { NgModule } from '@angular/core'; import {
BrowserModule } from '@angular/platform-browser';

import { AppComponent } from './app.component'; import {
ParentComponent } from './parent/parent.component'; import {
ChildComponent } from './child/child.component';

@NgModule({
  declarations: [
    AppComponent,
    ParentComponent,
    ChildComponent
  ],
  imports: [
    BrowserModule
  ],
  providers: [],   bootstrap:
[AppComponent]
})
export class AppModule {}
```

app.component.html :

```
<app-parent></app-parent>
```

parent.componet.ts:

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-parent',   templateUrl:
'./parent.component.html',   styleUrls:
['./parent.component.css']
})
export class ParentComponent {}
```

parent.component.html:

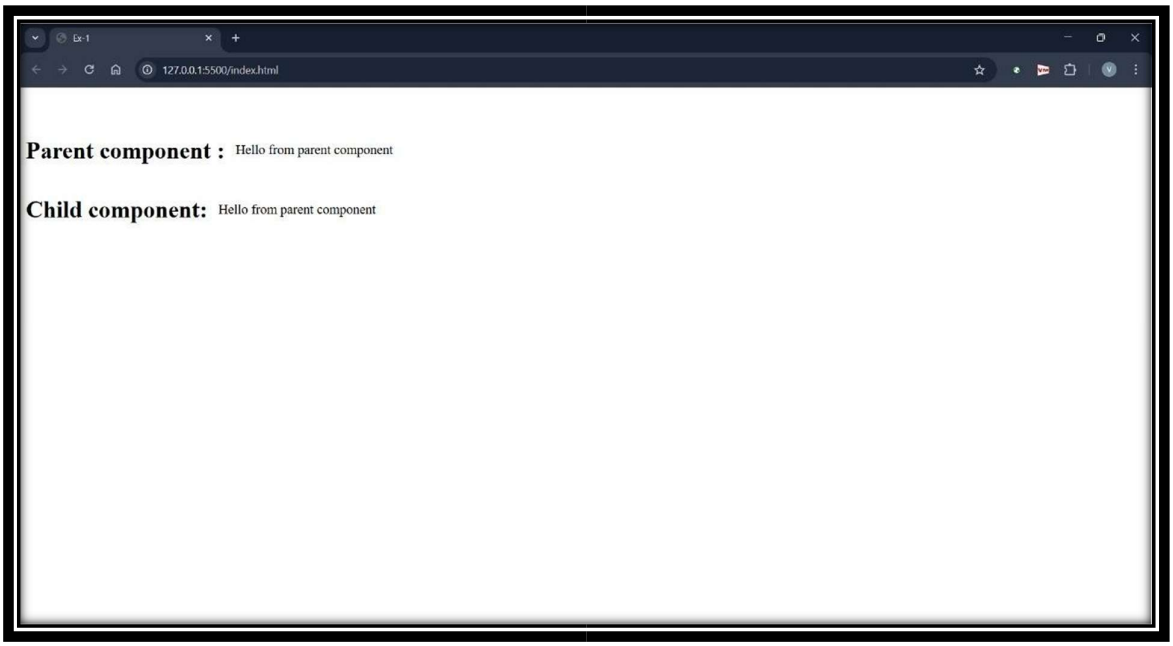
```
 <h1>This is the Parent Component</h1>
<app-child></app-child>
```

child.component.ts:

```
import { Component } from '@angular/core';
              @Component({
  selector: 'app-child',   templateUrl:
'./child.component.html',   styleUrls:
['./child.component.css']
}) export class ChildComponent
{}
```

```
child.componet.html: h1>This is the
Child Component</h1>
<app-parent></app-parent>
```

Output:


Ex-1
127.0.0.1:5500/index.html
Parent component : Hello from parent component
Child component: Hello from parent component

Program:

app.component.html:

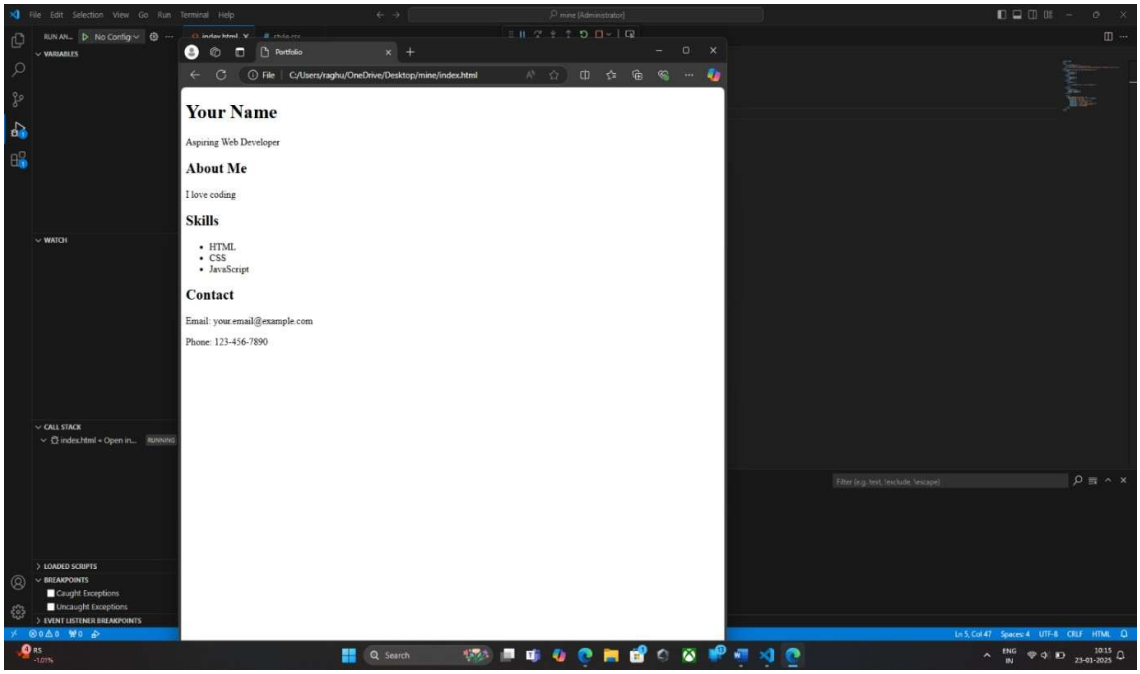
```
<!DOCTYPE html>
<html>
<head>
    <title>Portfolio</title>
    <link rel="stylesheet" href="css/style.css">
    <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
</head>
<body ng-app="portfolioApp" ng-controller="MainController">
    <header>
        <h1>{{name}}</h1>
        <p>{{role}}</p>
    </header>
    <section>
        <h2>About Me</h2>
        <p>{{about}}</p>
    </section>
    <section>
        <h2>Skills</h2>
        <ul>
            <li ng-repeat="skill in skills">{{skill}}</li>
        </ul>
    </section>
    <section>
        <h2>Contact</h2>
        <p>Email: {{email}}</p>
        <p>Phone: {{phone}}</p>
    </section>
```

```
    <script>
angular.module('portfolioApp', [])
        .controller('MainController', function ($scope) {
            $scope.name = "Your Name";
            $scope.role = "Aspiring Web Developer";
            $scope.about = "I love coding";
            $scope.skills = ["HTML", "CSS", "JavaScript"];
            $scope.email = "your.email@example.com";
            $scope.phone = "123-456-7890";
        });
    </script>
</body>
</html>
```

app.component.css: body {

```
font-family: Arial, sans-serif;
margin: 0;     padding: 0;
text-align: center;
} header {     background-
color: #333;     color:
white;     padding: 1rem 0;
} section {
margin: 1rem;
}

ul {     list-style:
none;     padding:
0; }
```

Output:


Portfolio
C:/Users/raghu/OneDrive/Desktop/mine/index.html
Your Name
Aspiring Web Developer
About Me
I love coding
Skills
HTML
CSS
JavaScript
Contact
Email: your.email@example.com
Phone: 123-456-7890

Program:

BACKEND:

Server.js:

```
const express = require("express"); const
bodyParser = require("body-parser"); const
path = require("path");

const app = express(); app.use(bodyParser.json());

// Serve static files (like index.html) from the current directory
app.use(express.static(path.join(__dirname)));

// In-memory storage for posts const
posts = [];

// API to create a post app.post("/post",
(req, res) => {
  posts.push({ author: req.body.author, content: req.body.content });
res.json({ message: "Post created" });
});
// API to get all posts (feed) app.get("/feed",
(req, res) => {   res.json(posts);
});

// Start the server app.listen(5000,
() => {
  console.log("Server running on http://localhost:5000");
});
```

FRONTEND:

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Micro-Blogging App</title>
</head>
<body>
  <h1>Micro-Blogging App</h1>

  <!-- Create a Post -->
  <h3>Create Post</h3>
  <input id="author" placeholder="Your name" />
  <input id="content" placeholder="Write something..." />
  <button onclick="createPost()">Post</button>

  <!-- Feed -->
  <h3>Feed</h3>
  <div id="feed"></div>

  <script>     const API_URL =
"http://localhost:5000";

    // Create a post     async
function createPost() {
```

```
    const author = document.getElementById("author").value;
const content = document.getElementById("content").value;

    await fetch(`${API_URL}/post`, {        method:
"POST",        headers: { "Content-Type":
"application/json" },        body: JSON.stringify({
author, content }),

    });
    // Reload the feed
loadFeed();
  }
  // Load the feed     async
function loadFeed() {
    const res = await fetch(`${API_URL}/feed`);
const posts = await res.json();

    document.getElementById("feed").innerHTML = posts
      .map(post => `<p><strong>${post.author}:</strong> ${post.content}</p>`)
      .join("");
  }
  // Load feed on page load
loadFeed();
 </script>
</body>
</html>
```

Output:

Micro-Blogging App

Create Post

Your name | Write something... | Post

Feed

john: hello world

rosy: heyy john

Program:

Home.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Home - Online Bookstore</title>
</head>
<body>
  <h1>Online Bookstore</h1>
  <nav>
    <a href="home.html">Home</a> |
    <a href="catalogue.html">Catalogue</a> |
    <a href="cart.html">Cart</a> |
    <a href="login.html">Login</a> |
    <a href="registration.html">Register</a>
  </nav>
  <p>Welcome to our bookstore! Find your next favorite book.</p>
</body>
</html>
```

Login.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Login - Online Bookstore</title>
</head>
<body>
  <h2>Login</h2>
  <nav>
```

```
    <a href="home.html">Home</a> |
    <a href="catalogue.html">Catalogue</a> |
    <a href="cart.html">Cart</a> |
    <a href="login.html">Login</a> |
    <a href="registration.html">Register</a>
  </nav>
  <form>
    <input type="text" placeholder="Username" required><br>
    <input type="password" placeholder="Password" required><br>
    <button>Login</button>
  </form>
</body>
</html>
```

Catalogue.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Catalogue - Online Bookstore</title>
</head>
<body>
  <h2>Book Catalogue</h2>
  <nav>
    <a href="home.html">Home</a> |
    <a href="catalogue.html">Catalogue</a> |
    <a href="cart.html">Cart</a> |
    <a href="login.html">Login</a> |
    <a href="registration.html">Register</a>
  </nav>
```

```
  <ul>
    <li>Book 1</li>
    <li>Book 2</li>
    <li>Book 3</li>
  </ul>
</body>
</html>
```

Cart.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Cart - Online Bookstore</title>
</head>
<body>
  <h2>Your Cart</h2>
  <nav>
    <a href="home.html">Home</a> |
    <a href="catalogue.html">Catalogue</a> |
    <a href="cart.html">Cart</a> |
    <a href="login.html">Login</a> |
    <a href="registration.html">Register</a>
  </nav>
  <p>No items in your cart yet!</p>
</body>
</html>
```

Registration.html:

```
<!DOCTYPE html>
<html>
```

```
<head>
  <title>Register - Online Bookstore</title>
</head>
<body>
  <h2>Register</h2>
  <nav>
    <a href="home.html">Home</a> |
    <a href="catalogue.html">Catalogue</a> |
    <a href="cart.html">Cart</a> |
    <a href="login.html">Login</a> |
    <a href="registration.html">Register</a>   </nav>
  <form>
    <input type="text" placeholder="Full Name" required><br>
    <input type="email" placeholder="Email" required><br>
    <input type="password" placeholder="Password" required><br>
    <button>Register</button>
  </form>
</body>
</html>
```

Output:

Login
Home | Catalogue | Cart | Login | Register
Username
Password
Login

Program:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Shopping Items Application</title>
   <script type="text/javascript"
   src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
<script src="app.js"></script>
</head>
<body ng-app="myApp">
<div ng-controller="myCntrl">
   <h2>Shopping Application</h2>
   <h4>List of Shopping Items</h4>
   <table border="1">
      <tr>
         <th>SLNO</th>
         <th>Item</th>
      </tr>
      <tr ng-repeat="item in shoppingItems">
         <td>{{$index + 1}}</td>
         <td>{{item}}</td>
      </tr>
   </table>
<br/>
```

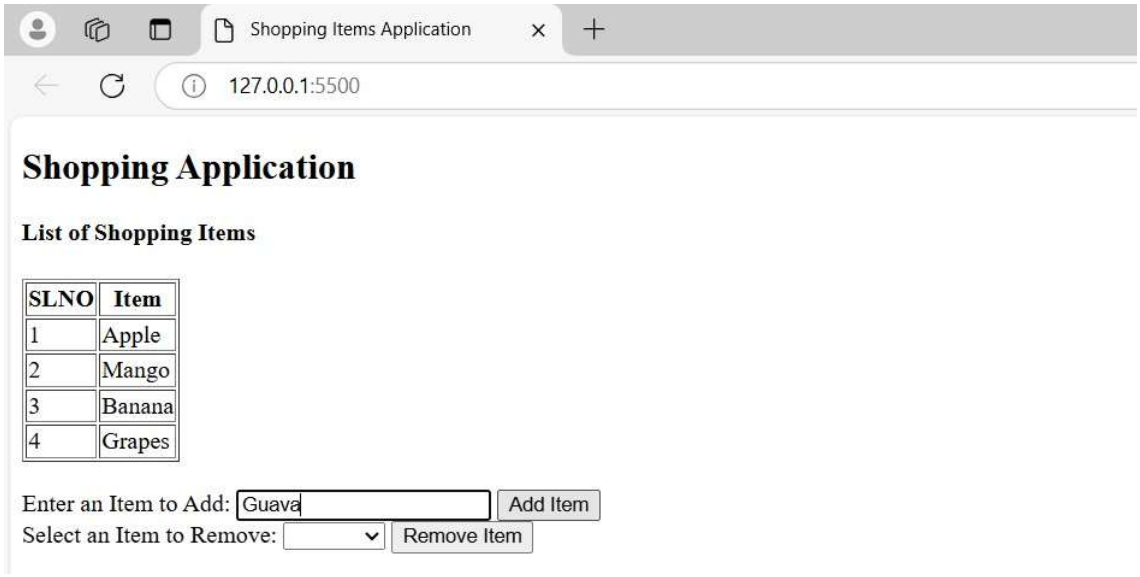
```
  <div>
      Enter an Item to Add: <input type="text" ng-model="newItem">
      <button ng-click="addItem()">Add Item</button>
   </div>

   <div>
      Select an Item to Remove:
      <select ng-model="selectItem" ng-options="item for item in shoppingItems"></select>
      <button ng-click="removeItem()">Remove Item</button>
   </div>
</div>  </body>
</html>
```

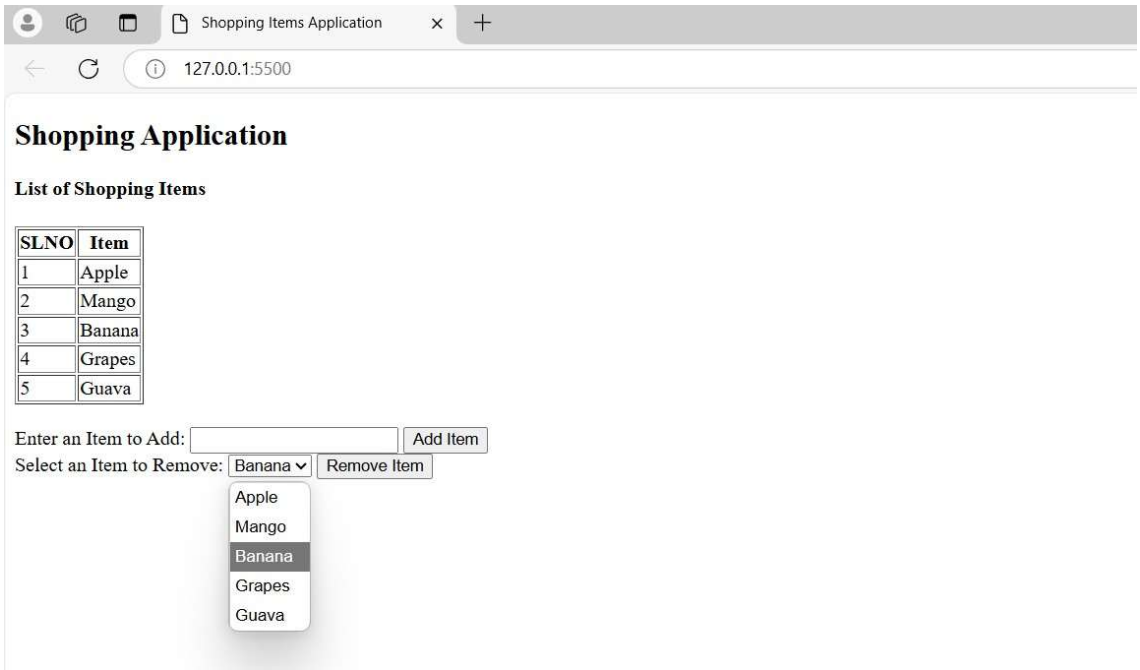
app.js:

```
var app = angular.module("myApp", []);  app.controller("myCntrl",
function($scope) {
   // Initial Shopping List Items
   $scope.shoppingItems = ['Apple', 'Mango', 'Banana', 'Grapes'];
 // Function to Add an Item     $scope.addItem = function() {          if ($scope.newItem
&& $scope.shoppingItems.indexOf($scope.newItem) === -1) {
          $scope.shoppingItems.push($scope.newItem);
          $scope.newItem = "";
       } else {
          if ($scope.newItem) {
             alert("This item is already in the shopping list.");
          } else {                alert("Please enter
an item to add.");
          }
       }
   };
```

```
    // Function to Remove an Item     $scope.removeItem = function()
{         if ($scope.shoppingItems.indexOf($scope.selectItem) === -
1) {             alert("Please select an item to remove.");
        } else {
            var index = $scope.shoppingItems.indexOf($scope.selectItem);
            $scope.shoppingItems.splice(index, 1);
            $scope.selectItem = "";
        }
    };
});
```

Output:


Shopping Items Application
127.0.0.1:5500
Shopping Application
List of Shopping Items
SLNO
Item
1
Apple
2
Mango
3
Banana
4
Grapes
Enter an Item to Add:
Guava
Add Item
Select an Item to Remove:
Remove Item


Shopping Items Application
127.0.0.1:5500
Shopping Application
List of Shopping Items
SLNO
Item
1
Apple
2
Mango
3
Banana
4
Grapes
5
Guava
Enter an Item to Add:
Add Item
Select an Item to Remove:
Banana
Remove Item
Apple
Mango
Banana
Grapes
Guava

Program:

Jsx file: import React, { useState }

```
from 'react'; import './Calculator.css';


const Calculator = () => {   const [display,
setDisplay] = useState('0');


  const handleNumber = (num) => {
    if (display === '0') {
setDisplay(num);     } else {
setDisplay(display + num);
    }
  };


  const handleOperator = (op) => {
const lastChar = display.slice(-1);
const operators = ['+', '-', '*', '/'];


    if (operators.includes(lastChar)) {
setDisplay(display.slice(0, -1) + op);
    } else {
setDisplay(display + op);
    }
  };


  const handleDecimal = () => {     const parts =
display.split(/[+\-*/]/);     const currentNumber =
parts[parts.length - 1];
```

```
    if (!currentNumber.includes('.')) {
setDisplay(display + '.');

    }
  };

  const calculate = () => {
    try {       const result =
eval(display);
setDisplay(result.toString());
    } catch (error) {
setDisplay('Error');
    }
  };

  const clear = () => {
setDisplay('0');
  };

  return (
    <div className="calculator">
      <div className="display" data-testid="display">{display}</div>
      <div className="buttons">
        <button className="clear" onClick={clear}>C</button>
<button onClick={() => handleOperator('/')}>/</button>
        <button onClick={() => handleOperator('*')}>*</button>
        <button onClick={() => handleNumber('7')}>7</button>
        <button onClick={() => handleNumber('8')}>8</button>
        <button onClick={() => handleNumber('9')}>9</button>
        <button onClick={() => handleOperator('-')}>-</button>
```

```
        <button onClick={() => handleNumber('4')}>4</button>
        <button onClick={() => handleNumber('5')}>5</button>
        <button onClick={() => handleNumber('6')}>6</button>
        <button onClick={() => handleOperator('+')}>+</button>
        <button onClick={() => handleNumber('1')}>1</button>
        <button onClick={() => handleNumber('2')}>2</button>
        <button onClick={() => handleNumber('3')}>3</button>
        <button className="equals" onClick={calculate}>=</button>
        <button className="zero" onClick={() => handleNumber('0')}>0</button>
        <button onClick={handleDecimal}>.</button>
      </div>
    </div>
  );
};
```

Css file: .calculator { width: 320px;
margin: 50px auto; padding: 20px;
background-color: #f5f5f5; border-
radius: 10px; box-shadow: 0 0 10px
rgba(0,0,0,0.2);
}

.display { background-
color: #fff; padding:
20px; margin-bottom:
20px; border-radius: 5px;
text-align: right; font-size:
2em; min-height: 60px;
}

```
.buttons {   display: grid;   grid-
template-columns: repeat(4, 1fr);
gap: 10px;

}

button {   padding: 20px;   font-
size: 1.2em;   border: none;
border-radius: 5px;   cursor:
pointer;   background-color:
#e0e0e0;   transition: background-
color 0.3s;
}

button:hover {   background-
color: #d0d0d0;
}
.clear {   background-color:
#ff6666;   color: white;
}

.equals {   background-color:
#4CAF50;   color: white;
grid-row: span 2;
}

.zero {   grid-column:
span 2;
}
```

Output:

Program:

JSX file: import React, { useState }

from 'react';

```
const VotingApp = () => {   const [candidates,
setCandidates] = useState([
    { id: 1, name: 'Candidate 1', votes: 0 },
    { id: 2, name: 'Candidate 2', votes: 0 },
{ id: 3, name: 'Candidate 3', votes: 0 },
  ]);

  const handleVote = (candidateId) => {
setCandidates(prevCandidates =>
prevCandidates.map(candidate =>
candidate.id === candidateId
          ? { ...candidate, votes: candidate.votes + 1 }
          : candidate
      )
    );
  };

  const totalVotes = candidates.reduce((sum, candidate) => sum + candidate.votes, 0);

  return (
    <div style={styles.container}>
      <h1 style={styles.title}>Voting System</h1>
      <div style={styles.total}>Total Votes: {totalVotes}</div>

      <div style={styles.candidatesContainer}>
```

```
        {candidates.map(candidate => (
          <div key={candidate.id} style={styles.candidateCard}>
            <h3>{candidate.name}</h3>
            <p>Votes: {candidate.votes}</p>
            <button                style={styles.button}
onClick={() => handleVote(candidate.id)}
            >
              Vote
            </button>
          </div>
        ))}
      </div>
    </div>
  );
};
```

Css file: const styles = {

```
container: {     maxWidth:
'800px',     margin: '0 auto',
padding: '20px',     fontFamily:
'Arial, sans-serif',
  },   title: {
textAlign: 'center',
color: '#333',   },
total: {     textAlign:
'center',     fontSize:
'1.2em',     margin:
'20px 0',
fontWeight: 'bold',
```

```
  },
  candidatesContainer: {
    display: 'flex',
    justifyContent: 'space-around',
flexWrap: 'wrap',
  },
  candidateCard: {     border: '1px solid
#ddd',     borderRadius: '8px',     padding:
'20px',     margin: '10px',     width: '200px',
textAlign: 'center',     boxShadow: '0 2px
4px rgba(0,0,0,0.1)',
  },   button: {
backgroundColor: '#4CAF50',
    color: 'white',
padding: '10px 20px',
border: 'none',
borderRadius: '4px',
cursor: 'pointer',
fontSize: '16px',
marginTop: '10px',
  },
};
```

Output:

Voting System
Total Votes: 8
Candidate 1
Votes: 5
Vote
Candidate 2
Votes: 2
Vote
Candidate 3
Votes: 1
Vote

Program:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>User Management</title>
   <script>        const apiUrl =
"http://localhost:5000/users";


       // Fetch and display users
function fetchUsers() {
fetch(apiUrl)
              .then(response => response.json())
              .then(users => {                    const userList =
document.getElementById("userList");                    userList.innerHTML = "";
users.forEach(user => {                    const li = document.createElement("li");
li.innerHTML = `${user.name} (${user.email}) - ${user.age} years
<button onclick="deleteUser('${user._id}')">Delete</button>`;
userList.appendChild(li);
                 });
              })
              .catch(error => console.error("Error fetching users:", error));
       }
       // Add a new user        function addUser() {            const
name = document.getElementById("name").value;            const
email = document.getElementById("email").value;            const
```

```
age = document.getElementById("age").value;
fetch(apiUrl, {
            method: "POST",              headers: { "Content-Type":
"application/json" },              body: JSON.stringify({ name, email,
age: Number(age) })
        })
        .then(response => response.json())
        .then(() => {
            fetchUsers(); // Refresh user list
document.getElementById("name").value = "";
document.getElementById("email").value = "";
document.getElementById("age").value = "";
        })
        .catch(error => console.error("Error adding user:", error));
    }
    // Delete a user        function deleteUser(id) {
fetch(`${apiUrl}/${id}`, { method: "DELETE" })
            .then(() => fetchUsers())
            .catch(error => console.error("Error deleting user:", error));
    }

    document.addEventListener("DOMContentLoaded", fetchUsers);
  </script>
</head>
<body>
  <h2>User Management</h2>
  <div>
    <input type="text" id="name" placeholder="Name">
    <input type="email" id="email" placeholder="Email">
```

```
        <input type="number" id="age" placeholder="Age">
        <button onclick="addUser()">Add User</button>
    </div>
    <h3>Users:</h3>
    <ul id="userList"></ul>
</body>
</html>
```

Server.js:

```
require("dotenv").config(); const express =
require("express"); const mongoose =
require("mongoose"); const bodyParser =
require("body-parser"); const cors =
require("cors");

const app = express();

// Middleware
app.use(cors());
app.use(bodyParser.json()); //
Connect to MongoDB
mongoose
  .connect(process.env.MONGO_URI, { useNewUrlParser: true, useUnifiedTopology: true ,
family: 4})
  .then(() => console.log("MongoDB Connected"))
  .catch((err) => console.error("MongoDB Connection Error:", err));
// Define a User Schema const UserSchema =
new mongoose.Schema({   name: String,
email: String,   age: Number,
});
const User = mongoose.model("Schema", UserSchema);
```

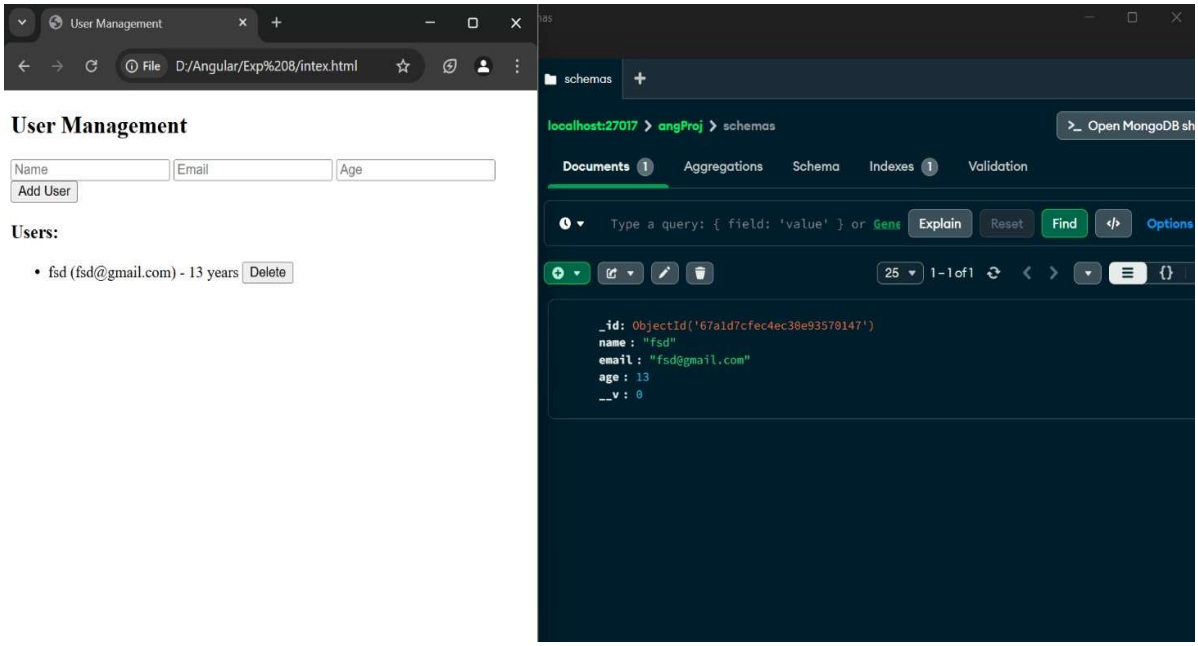
```
// Routes app.get("/", (req,
res) => {
res.send("Welcome to the
MongoDB Node.js API!");

});
// Create a User app.post("/users",
async (req, res) => {     try {
     const newUser = new User(req.body);
const savedUser = await newUser.save();
console.log("User saved:", savedUser);
res.status(201).json(savedUser);
    } catch (error) {
     console.error("Error saving user:", error);
res.status(400).json({ error: error.message });
    }
  });
// Get All Users app.get("/users",
async (req, res) => {   const users =
await User.find();   res.json(users);
});
// Get a Single User app.get("/users/:id", async
(req, res) => {   try {     const user = await
User.findById(req.params.id);
    if (!user) return res.status(404).json({ message: "User not found" });
res.json(user);   } catch (error) {     res.status(400).json({ error:
error.message });
  }
});
```

```
// Update a User app.put("/users/:id",

async (req, res) => {   try {

   const updatedUser = await User.findByIdAndUpdate(req.params.id, req.body, {      new:
true,

   });

   if (!updatedUser) return res.status(404).json({ message: "User not found" });

res.json(updatedUser);   } catch (error) {     res.status(400).json({ error:

error.message });

  }

});

// Delete a User app.delete("/users/:id",

async (req, res) => {   try {

   const deletedUser = await User.findByIdAndDelete(req.params.id);     if

(!deletedUser) return res.status(404).json({ message: "User not found" });

res.json({ message: "User deleted successfully" });

  } catch (error) {     res.status(400).json({

error: error.message });

  }

});

// Start Server const PORT = process.env.PORT || 5000; app.listen(PORT,

() => console.log(`Server running on port ${PORT}`));
```

.env file:

```
MONGO_URI="mongodb://localhost:27017/”
```

Output:


User Management
File D:/Angular/Exp%208/intex.html
User Management
Name
Email
Age
Add User
Users:
fsd (fsd@gmail.com) - 13 years Delete
schemas
localhost:27017 > angProj > schemas
Open MongoDB sh
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Gene
Explain
Reset
Find
Options
25
1-1 of 1
_id: ObjectId('67a1d7cfec4ec30e93570147')
name : "fsd"
email : "fsd@gmail.com"
age : 13
__v : 0